Survey interviews across three waves

Two independent NORC extracts support six descriptive questions about US adults. The usable sample depends on which questions each interview contained.

Why these years

A includes 2014 and 2016. B includes 2016 and 2018. These consecutive waves give roughly 5,000 interviews per extract, retain coverage of every requested concept and supply a shared wave for testing duplicate handling. The same years serve all questions. Each query defines its own eligible population.

Wave	Cut A	Cut B	Distinct interviews
2014	2,538	-	2,538
2016	2,867	2,867	2,867
2018	-	2,348	2,348
All waves	5,405	5,215	7,753

All 2,867 shared interviews agree field for field. They comprise 53.04% of A and 54.98% of B. The 2012-2016 alternative gave fewer observations for web duration. The 2016-2021 alternative introduced survey mode and wording changes in 2021. Each would still meet the overlap requirement.

What the files contain

Explorer supplied 32 fields and an embedded codebook. Each archive contained a strict JSON file with aligned column arrays and another file formatted as a Python literal. Both representations produced identical records. The repository preserves the strict JSON bytes and records their hashes.

Usable answers / interviews	2014	2016	2018
EVWORK / past employment	887	1,023	816
SATJOB / work satisfaction	1,895	1,779	1,739
TVHOURS / daily TV time	1,669	1,883	1,555
INTTECH / technology interest	1,238	1,386	1,173
Both web time components	1,398	1,587	1,362

The pre-build profile in docs/profile.csv lists missing shares for all 32 fields in all three years. The analyses are unweighted descriptions of the responding sample. The source weights are available for survey-aware follow-up.

What each answer means

All concepts have observations in 2014, 2016 and 2018. Having a field in a wave does not mean every respondent received that question.

Question	Fields and interpretation
Working now	WRKSTAT. Full time, part time and temporarily absent from a job count as working. Unknown status remains unknown.
Worked a year	EVWORK asks about working as long as one year. It is routed to retired people, students, homemakers and the other nonworking category. Duration for current workers and job seekers cannot be recovered.
Education	DEGREE supplies comparable credentials for industry groups. EDUC measures years of schooling.
Industry	INDUS10 identifies the industry. Restrict to working respondents because nonworkers can report an earlier industry.
Work satisfaction	SATJOB asks about work. Restrict the TV comparison to workers. In 2016 ballot A, form Y uses another wording; keep it outside the standard item.
Afterlife	POSTLIFE separates yes/no belief from uncertainty, no answer and unavailability.
Prayer	PRAY is an ordered frequency. Report the distribution within upbringing groups.
Raised religion	RELIG16 measures upbringing. RELIG is current preference; ATTEND and RELITEN describe attendance and affiliation strength.
Newspaper	NEWS measures reading frequency. Compare daily readership as a share of valid answers on ballots A and B.
Television	TVHOURS measures hours per day on ballots A and B. Zero hours contributes zero to the mean.
Technology	INTTECH measures interest in new inventions and technology issues. Its observed ballot/form coverage changes by year.

Media routing and other fields

WWWHR plus WWWMIN / 60 measures weekly non-email web time only when both components are valid. COMPUSE, WEBMOB, EMAILHR, EMAILMIN and USEWWW describe routing. A skipped USEWWW does not establish zero use. AGE, SEX, REGION, FORM, BALLOT, MODE and two person weights preserve context.

Valid INTTECH appears on B/X, B/Y and C/Y in 2014, and A/X, A/Y and B/X in 2016 and 2018. Ballot C accounts for most media omissions. One 2016 NEWS inapplicable response was outside ballot C, so its exact cause remains unresolved.

What the seven queries show

Percentages use observed, eligible responses. Each substantive query reads the respondent table, whose (year, ID) grain is tested against the input membership.

1 Prayer by religion raised in

7,659 interviews have both responses. Several-times-a-day prayer was reported by 1,413/4,007 raised Protestant (35.26%), 601/2,515 raised Catholic (23.90%) and 28/49 raised Muslim (57.14%). Small groups are particularly sensitive to a few responses. The full frequency distribution and denominators are in 01_prayer.csv.

2 Belief in life after death

Yes responses were 1,783/2,242 (79.53%) in 2014, 2,089/2,588 (80.72%) in 2016 and 1,720/2,123 (81.02%) in 2018. Each wave had zero recorded not-asked cases. Excluded uncertain/no-answer counts were 296, 279 and 225 respectively.

3 Most frequent industry among workers

Construction led for less than high school (53/384) and high school (186/2,275). Hospitals led for associate degrees (41/398). Elementary and secondary schools led for bachelor's (109/1,037) and graduate degrees (94/582). These are pluralities among workers with known credentials and industry.

4 TV hours by job satisfaction

Satisfaction	Very satisfied	Moderately satisfied	A little dissatisfied	Very dissatisfied
Group size	1,415	1,091	251	88
Mean TV hours/day	2.32	2.37	2.39	2.55

The extreme groups differ by 0.23 hours a day. The smallest group has 88 people. The comparison is descriptive and does not account for age, hours worked or survey design.

5 Ever worked at least one year

Among 3,034 nonworkers, 2,728 were eligible and 2,726 answered. Of those, 2,467 said yes (90.50%). For all nonworkers, assigning every unknown to no or to yes gives bounds of 81.31%-91.46%. The point estimate for all nonworkers is unidentified.

6 Changes between waves

Daily newspaper reading fell from 24.90% to 21.11% to 20.01% (changes of -3.78 and -1.10 percentage points, calculated before rounding). TV hours/day averaged 2.98, 3.03 and 2.94. These compare different samples at each wave.

7 Overlap audit

5,405 + 5,215 - 2,867 = 7,753 distinct interviews. The audit uses both file memberships. Results 1-6 use a single copy of each interview.

Operation, cost and oversight

Python validates and loads files. DuckDB holds raw rows and analytical tables in one local database. dbt makes the transformations, definitions and tests inspectable.

From source to respondent

A source row belongs to one file and ordinal. A respondent key is the survey year plus the within-year ID. Three materialized models store membership keys, decoded answers and one row per interview. Versioned seeds describe each variable, its domain, reviewed years, source codes and missing types. Genuine zero, no answer, uncertainty, inapplicability, absent property and year-unavailable responses remain distinguishable. Original labels stay accessible.

Reruns and failure

File hashes skip unchanged inputs. Conflicting overlap or malformed input rolls back the load. New files enter raw storage once. The next dbt build regenerates the small downstream tables. Test failure stops documentation and results. The run log records hashes, counts, stages and exceptions. Local tests compared actual A+B and B+A, repeated each load, and checked rollback on a deliberate conflict.

Measured change and scale

The initial membership view parsed keys from JSON on every audit. Screening all seven queries identified the overlap audit as slowest. Materializing the four membership columns reduced the median from 90.87 to 7.97 ms on this machine, single thread, one warm-up and three timed runs. Both versions returned the same result. The key-table build took about 0.51 s in dbt, including model overhead. The plans and timings are in query-plans.txt.

The 30 non-key variables produce 232,590 answer rows today. At 75,700 interviews and 7,000 variables the dense expansion approaches 530 million rows. Full rebuilding, JSON parsing and memory for whole input arrays would become the first constraints. Project only needed variables, store typed columnar partitions, and rebuild affected waves when measured costs warrant it. Keep versioned semantics per source. A local runner and pinned virtual environment suffice for these manual batches. PostgreSQL and Airflow would add two services to operate for a job that runs on demand.

Contribution, limitations and next checks

Codex assisted with year comparisons, profiling, design, implementation, SQL, testing and report drafting. The candidate approved the year-selection plan, requested a simpler scope and supplied the two Explorer archives. Automated checks were run by Codex. Candidate review of individual findings and personal hours require the candidate's own account.

About ten minutes elapsed between the profile/design commit and the first verified report in this local build session. This is a partial effort measure: earlier research, export preparation and the personal effort are outside that interval. With more time, verify ambiguous routing against the questionnaires and assess weighted estimates and uncertainty before drawing population conclusions.